A DEVELOPER'S REFLECTION

I Built a Memory System for AI Agents. It Started With Markdown Files.

30 days, 700+ commits, and the tool that learned to build itself.

Rust | Open Source | AI Memory | MCP | Building in Public

THE WHY

Why Build This?

🧠 Learn about generative AI firsthand. Not by reading. By building in it and feeling the impact directly.

🧰 Own my development tool. Every role change, I waste time rebuilding my environment. I wanted something portable.

📂 Build a portfolio that ships. Not a tutorial project. A real product I'd use every day.

PERSONAL MEMORY

"use tabs not spaces in this project"

CLAUDE.md, Cursor rules, Copilot

ORGANIZATIONAL MEMORY

"auth was redesigned across 3 repos last month. Here are the trade-offs and what testing revealed."

corvia

THE PROBLEM

What's Missing

Three agents. One codebase. Zero shared memory.

ISOLATED

🤖 Agent A redesigns auth across 3 repos

🤖 Agent B finds a side effect during testing

❌ **Agent C starts next week. Knows nothing.**

CONNECTED

🤖 Agent A redesigns auth → recorded

🤖 Agent B finds side effect → recorded

✅ **Agent C queries first. Already knows both.**

That's the difference between personal memory and organizational memory. corvia builds the second layer.

THE BOOTSTRAP

Day 1: No Tool Yet

Building a memory system that can't remember anything. Yet.

Markdown files

Asked Claude to save every decision in a specific format. Brainstorms, trade-offs, architecture calls. All by hand. The plan: migrate it all once the tool is ready.

Obra's superpowers plugin

The brainstorming skill structured every design session. Explore, clarify, converge. This was the process from day 1. The rest came later.

First self-ingest

corvia ingests its own decisions. The markdown files migrate in. Now the tool can search its own history. The loop starts forming.

The loop gets better

5-persona review, session history, agent coordination. The process and the product grew together.

THE PROCESS

The Loop Evolved

Day 1 vs. today.

Week 1

1. Brainstorm + Design
2. Build
3. Try to record decisions

(some got missed)

Today

1. Search what's decided
2. Brainstorm + Design
3. Build + Test
4. 5-Persona Review
5. Record into corvia

The process and the product co-evolved. Each capability added to corvia improved how the next one got built.

THE REAL TEST

Beyond Dogfooding

Interview take-home

First time using corvia outside its own development. Structured decision recording. Querying past decisions before building. I didn't expect it to be ready.

< 24 hours

The methodology transferred. That was the moment it stopped being just a portfolio project.

The honest part

Not everything shipped perfectly. A lot of recent features haven't been fully used yet. Agent teams, structural code intelligence, session history. Built and tested, but I'm still internalizing them.

Getting hands-on with AI development is fundamentally different from reading about it. You have to feel it break, watch it recover, and build intuition from there.

THE MINDSET

Staying on Edge

When the space moves, move with it.

Claude ships memory, Agent Teams, session persistence
Personal memory becomes table stakes.
→ Exciting, not threatening. The organizational layer is still open.

Claude Agent Teams launches
Teams of agents solving problems together.
→ Built an adapter. How can corvia capture what teams discover?

GitNexus shows structural intelligence
Most codebases aren't started from scratch.
→ Started building structural code analysis into corvia's adapter.

AI is moving fast. Grab every emerging capability and build on it. That's how you stay on edge.

BY THE NUMBERS

30 Days of Building

February 28 to March 29, 2026.

700+
Commits (both repos)

12
Releases (v0.3.1 to v0.4.6)

1,100+
Tests passing

100+
Decisions recorded

10
Rust crates in workspace

21
MCP tools shipped

CURRENT STATE

What Exists Today

Four pillars. Sixteen capabilities.

🔍

Knowledge

Semantic search
Knowledge graph
Bi-temporal queries
Automated reasoning

What do we know? When did we know it?

🤝

Agents

Multi-agent coordination
Crash recovery
Agent teams
Session history

Multiple agents. No collisions. No data loss.

⚡

Infrastructure

GPU inference (CUDA + iGPU)
Built-in ONNX runtime
21 MCP tools
Adapter plugin system

Runs locally. Integrates via MCP.

📊

Observability

Web dashboard
OTEL traces
CLI metrics
Docs workflow

See what's happening inside.

corvia
The product. Kernel, server, CLI, adapters. 10 Rust crates.

corvia-workspace
The lab. Wraps repos, holds the knowledge store, runs the devcontainer.

KEY TAKEAWAYS

What I Learned

🔁 **Dogfood as early as possible**

Race to self-host. The sooner the tool builds itself, the sooner the loop reveals what's actually needed.

🖐️ **Hands-on beats reading**

You can't understand AI development from blog posts. You have to build, feel it break, and build intuition firsthand.

🌊 **Grab every emerging capability**

Claude shipped Agent Teams. I built an adapter. GitNexus showed structural intelligence. I started building that too. AI moves fast. Ride the wave.

🌐 **Personal memory is table stakes**

CLAUDE.md, Cursor rules, Copilot memory. What's still open is the organizational layer. Memory across repos, over time, between agents.

What's the messiest part of your AI development workflow?

The part you'd fix if you had time.

github.com/chunzhe10/corvia